



REPUBLIC OF SLOVENIA
MINISTRY OF THE ENVIRONMENT AND SPATIAL PLANNING
SLOVENIAN ENVIRONMENT AGENCY



Quality Assessment of Surface Current and Wave Measurements from the High-Frequency Radar System in the Gulf of Trieste

Section 2: **Evaluation of the effectiveness of existing quality control (QC) algorithms for measurement data**

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1 Introduction

For the HFR GOT measurement time series (2021 - 2026), the existing quality control (QC) procedures and algorithms implemented within the WERA system will be or are being used.

The aim of this section is to systematically evaluate their effectiveness: what proportion of artifacts and anomalies they actually detect, which artifacts they identify, which types of errors remain undetected, and where the weaknesses of the current methods lie.

Note 1: All scripts and corresponding results are provided alongside this document.

2 Review and documentation of existing QC procedures and algorithms that are currently used or available for HFR GOT data (both raw measurement data and derived products)

The HFR-NAdr network publishes data through the EU HFR Node, which applies a standardized QC suite to all member radar systems before publication to ERDDAP. It is cited in the data's own 'qc_manual' global attribute under: "Recommendation Report 2 on improved common procedures for HFR QC analysis", (<https://dx.doi.org/10.25607/0BP-944>) or it can be obtained via repository (https://github.com/LorenzoCorgnati/European_standard_HFR_data_model).

Tables 1 and 2 in Section 4 ("The European common QC model for real-time HFR data") represent the test-by-test methodology.

QC is applied at two product levels:

- Radial product: (per-station: AURI, IZOL, PIRA, TRI1) QC runs on each station's own line of sight radial velocity measurements before combination.
- Total product: QC runs again after two or more radials are combined into a vector current field on the regular grid.

A vector can therefore be flagged at the radial stage, the total-combination stage, or both.

2.1 Radial Product

Variable	Description	What it checks	Threshold
AVRB_QC	Average radial bearing	This test is applicable only to DF systems. Data files from BF systems will have this variable filled with “good data” flag and the explanation in the comment attribute	Bearing range = $0.0 - 360.0^\circ$
CSPD_QC	Velocity threshold	This test labels radial vectors whose module is bigger than a max. velocity threshold with a “bad data” flag and radial vectors whose module is smaller than the threshold with a “good data” flag	max. velocity = $1.2 \frac{m}{s}$
MDFL_QC	Median filter	For each source vector, the median of all velocities within a radius of $< RCLim >$ and whose vector bearing (angle of arrival at site) is also within an angular distance of $< AngLim >$ degrees from the source vector’s bearing is evaluated. If the difference between the vector’s velocity and the median velocity is greater than a threshold, then the vector is labelled with a “bad data” flag, otherwise it is labelled with a “good data” flag	Distance limit = 5.0 km; velocity median difference threshold = $1.0 \frac{m}{s}$
OWTR_QC	Over-water	This test labels radial vectors that lie on land with a “bad data” flag and radial vectors that lie on water with a “good data” flag	not a numeric threshold
POSITION_QC	Position	Validates the vector’s grid coordinate is within the expected domain	No fixed numeric threshold
QCflag	Overall	Composite: “good data” if every applicable test passes	No fixed numeric threshold

Variable	Description	What it checks	Threshold
RDCT_QC	Radial count	Test labelling radial data having a number of velocity vectors bigger than the threshold with a “good data” flag and radial data having a number of velocity vectors smaller than the threshold with a “bad data” flag	min. number of radial vectors = 25
VART_QC	Variance threshold	This test labels radial vectors whose temporal variance is bigger than a max. threshold with a “bad data” flag and radial vectors whose temporal variance is smaller than the threshold with a “good data” flag (applicable only on BF systems)	max. variance = $1.0 \frac{m^2}{s^2}$

Table 1: Mandatory QC tests for radial velocity data (for HFR Nadr).

2.2 Total Product

Variable	Description	What it checks	Threshold
CSPD_QC	Velocity threshold	This test labels total vectors whose module is bigger than a max. velocity threshold with a “bad data” flag and total vectors whose module is smaller than the threshold with a “good data” flag	max. velocity = $1.0 \frac{m}{s}$
DDNS_QC	Data density threshold	This test labels total velocity vectors with a number of contributing radials bigger than the threshold with a “good data” flag and total velocity vectors with a number of contributing radials smaller than the threshold with a “bad data” flag	min. contributing radial velocities = 3
GDOP_QC	GDOP threshold	This test labels velocity vectors whose GDOP is bigger than a max. threshold with a “bad data” flag and the vectors whose GDOP is smaller than the threshold with a “good data” flag	max. GDOP = 2.0
POSITION_QC	Position	Same as above (Table 1)	Same as above (Table 1)
QCflag	Overall	Same as above (Table 1)	Same as above (Table 1)
VART_QC	Variance threshold	This test labels total vectors whose temporal variance is bigger than a max. threshold with a “bad data” flag and total vectors whose temporal variance is smaller than the threshold with a “good data” flag (applicable only on BF systems)	max. variance = $1.0 \frac{m^2}{s^2}$

Table 2: Mandatory QC tests for total velocity data (for HFR Nadr)

2.3 QC indicators

The quality control flags indicate the data quality of the data values in a file, following the ARGO QC flag scale.

Code	Meaning	Comment
0	unknown	no QC was performed
1	good data	all QC tests passed
2	probably good data	these data should be used with caution
3	potentially correctable bad data	these data are not to be used without scientific correction or re-calibration
4	bad data	data have failed one or more QC tests
5	value changed	data may be recovered after transmission error
6	-	not used
7	nominal value	the provided value is not measured but comes from a priori knowledge, e.g. instrument target depth
8	interpolated value	missing data may be interpolated from neighbouring data in space or time
9	missing value	

Table 3: Argo QC flag scale.

3 Application of the existing QC algorithms to the 2021–2026 time series and quantitative analysis of the results: number of flagged / removed points over time and space, by measurement type and test type

Note 2: At this point, no new QC logic was written. The flags already exist in the downloaded netCDF files (computed by the EU HFR Node before publication); this task only quantifies what they already do.

To analyse the existing QC algorithms, the script *qc_flag_analysis.py* was developed and run. It loads all five unified 2021-present datasets (Total + AURI/IZOL/PIRA/TRI1), and for every QC test variable, it computes:

- Overall flag histogram: *qc_flag_summary.csv*
- Monthly “bad fraction” (flags 3 and 4 as a % of QC-evaluated cells, flags 1-4): *qc_flag_monthly.csv* and *qc_monthly_*.png*

- Spatial “bad fraction” per grid cell across the full record: *qc-spatial_*.png*

3.1 Results

The pipeline only ever emits flag 1 (good) or 4 (bad). Flags 0, 2, 3, 5-9 never appear anywhere in the archive. The ARGO 0-9 scale is defined, but the data effectively uses it as a binary pass/fail system.

AVRB_QC, as documented in the manual and in Task 1, shows 0.0% bad data across all four radial stations, with every single cell having *flag* = 1. This confirms the documentation; this test effectively contributes nothing to our QC and should not be counted as “coverage” in the following analysis.

Radial product: the RDCT_QC accounts for the large majority (88-97%) of QCflag’s overall bad rate at stations PIRA, TRI1, and IZOL. At AURI, this share is much smaller:

Station	QCflag % bad	RDCT_QC % bad	Next-largest test
AURI	0.85%	0.35%	VART_QC 0.26%
IZOL	17.66%	17.10%	OWTR_QC 0.37%
PIRA	0.96%	0.84%	OWTR_QC 0.10%
TRI1	6.60%	5.82%	OWTR_QC 0.80%

Table 4: Percentage of bad data flagged by QC tests for each HFR station.

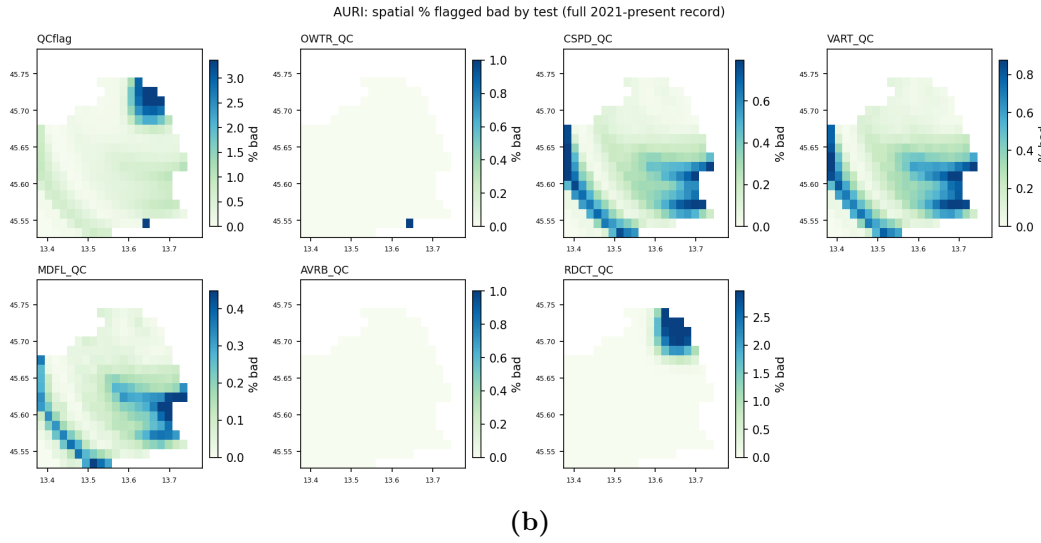
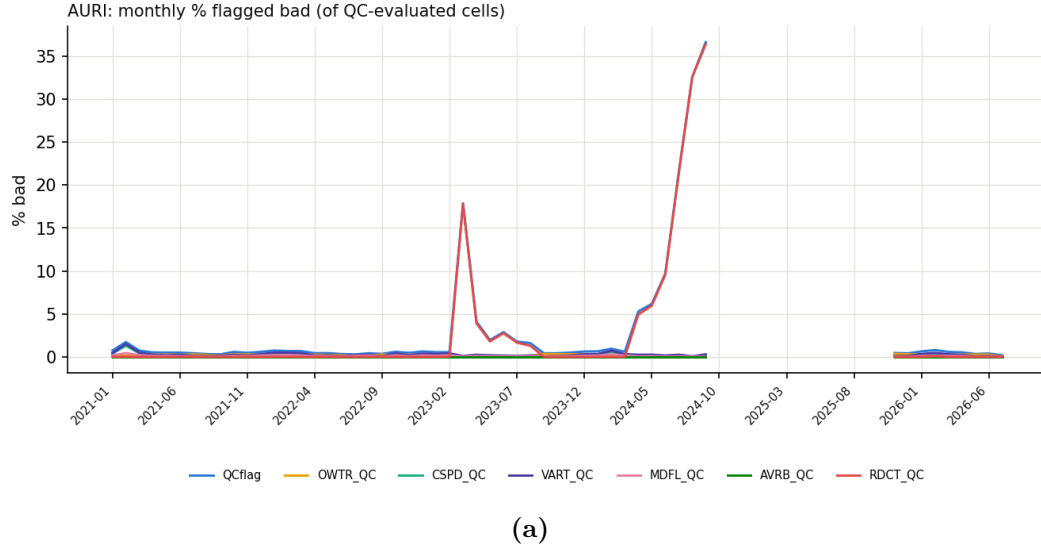
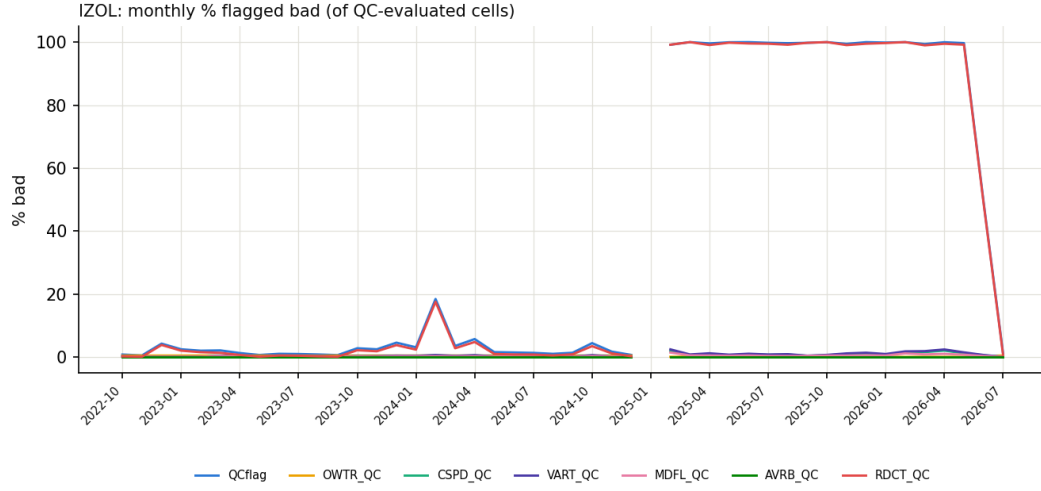
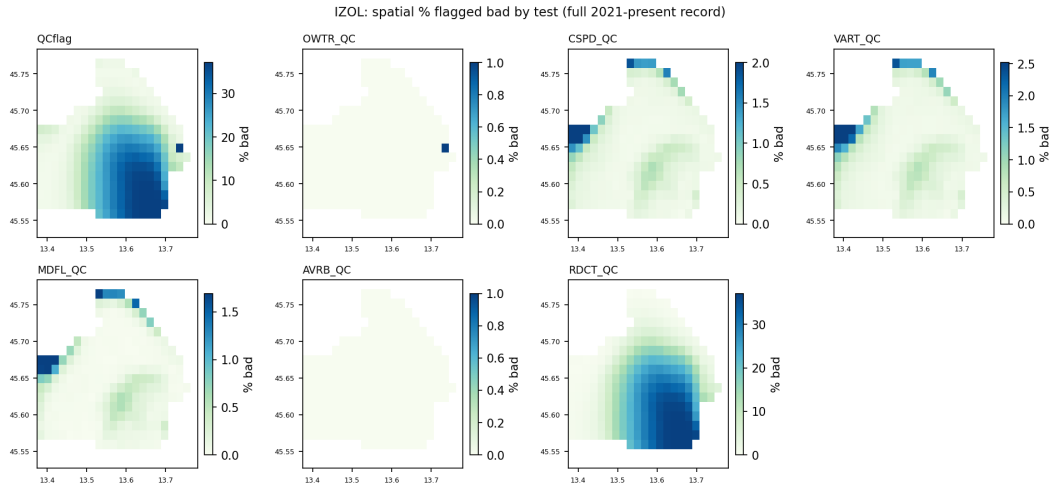


Figure 2: Quality-control results for the AURI station. (a) Monthly fraction of data flagged as bad by the QC tests. (b) Spatial distribution of the fraction of data flagged as bad over the full observation period.

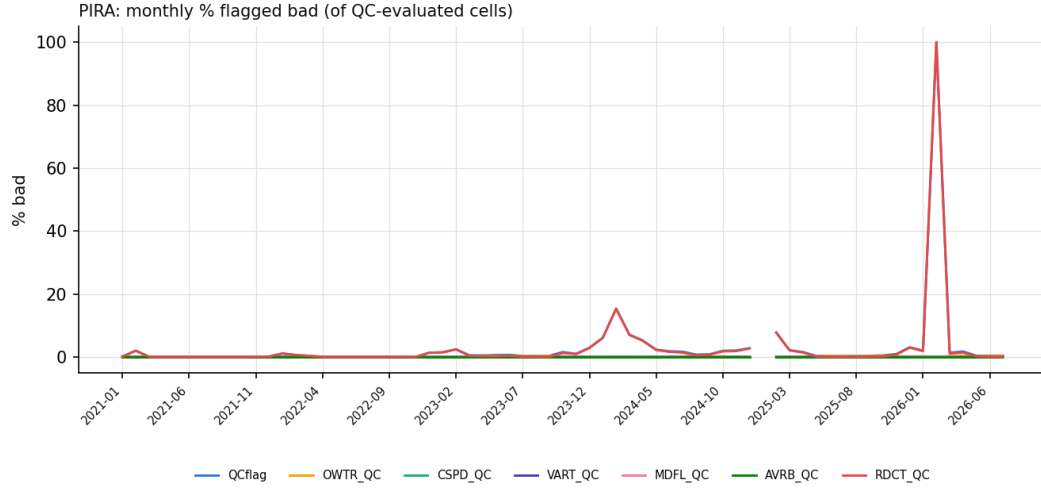


(a)

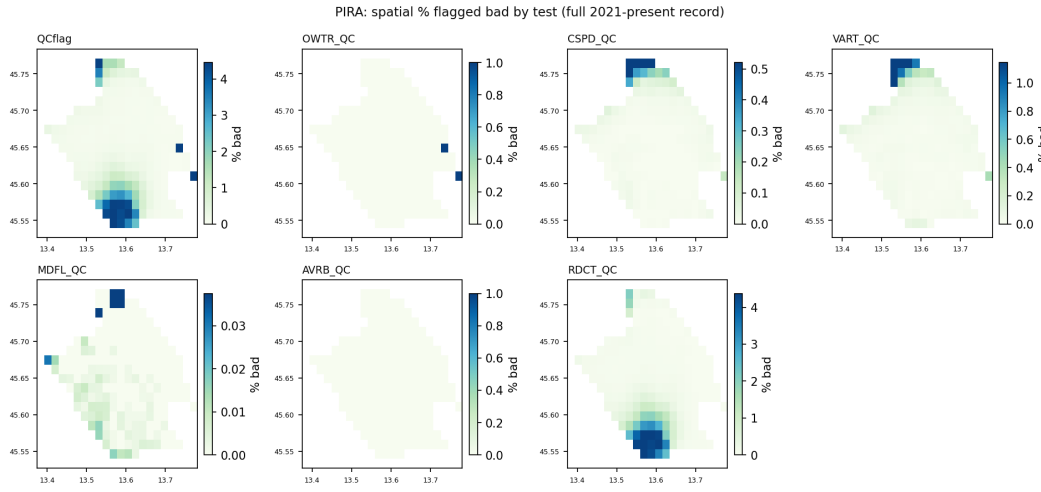


(b)

Figure 3: Quality-control results for the IZOL station. (a) Monthly fraction of data flagged as bad by the QC tests. (b) Spatial distribution of the fraction of data flagged as bad over the full observation period.

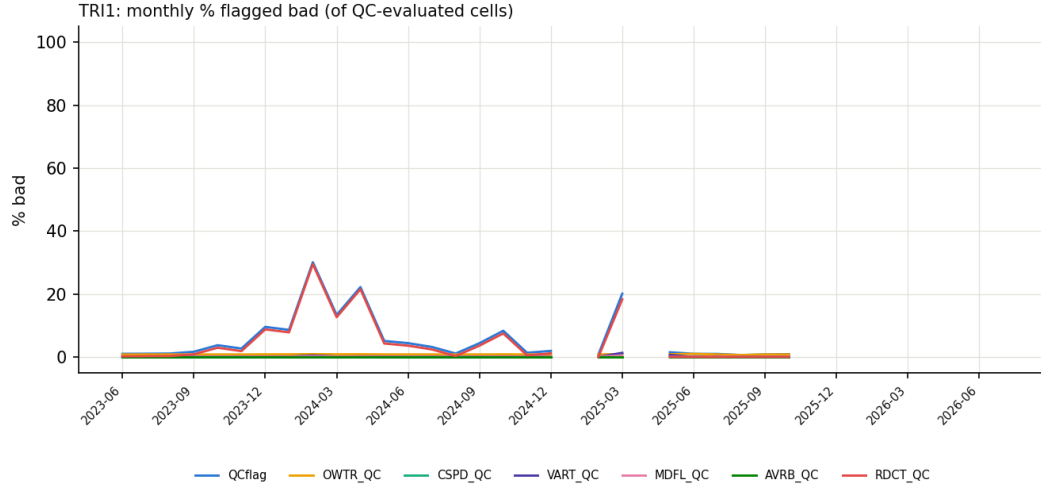


(a)

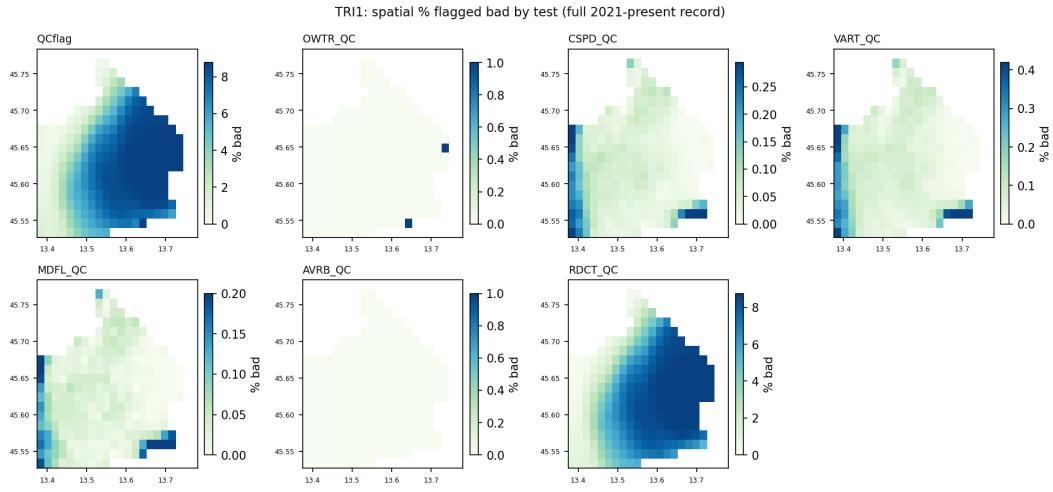


(b)

Figure 4: Quality-control results for the PIRA station. (a) Monthly fraction of data flagged as bad by the QC tests. (b) Spatial distribution of the fraction of data flagged as bad over the full observation period.



(a)



(b)

Figure 5: Quality-control results for the TRI1 station. (a) Monthly fraction of data flagged as bad by the QC tests. (b) Spatial distribution of the fraction of data flagged as bad over the full observation period.

Total product: the GDOP_QC is the dominant failure mode (18.6% bad, vs. CSPD_QC 0.24%, VART_QC/DDNS_QC 0.0%).

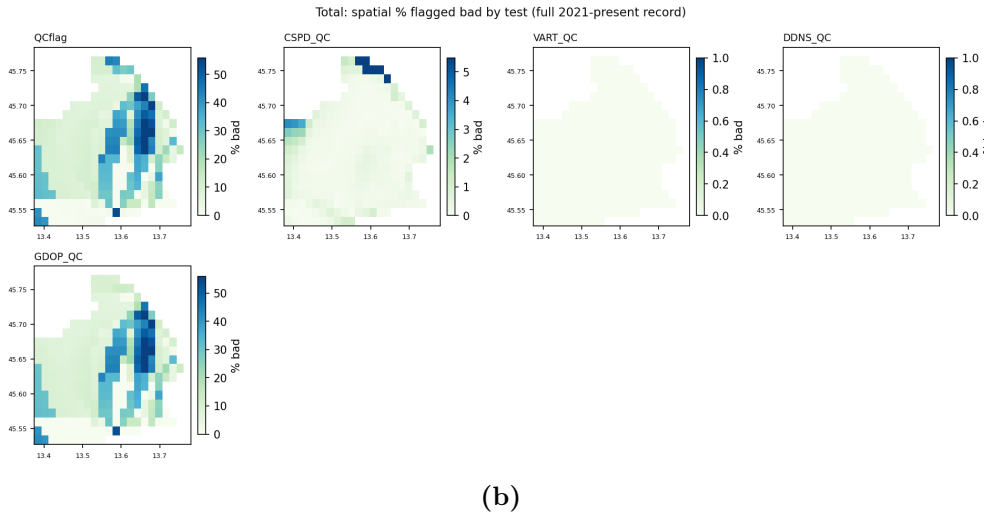
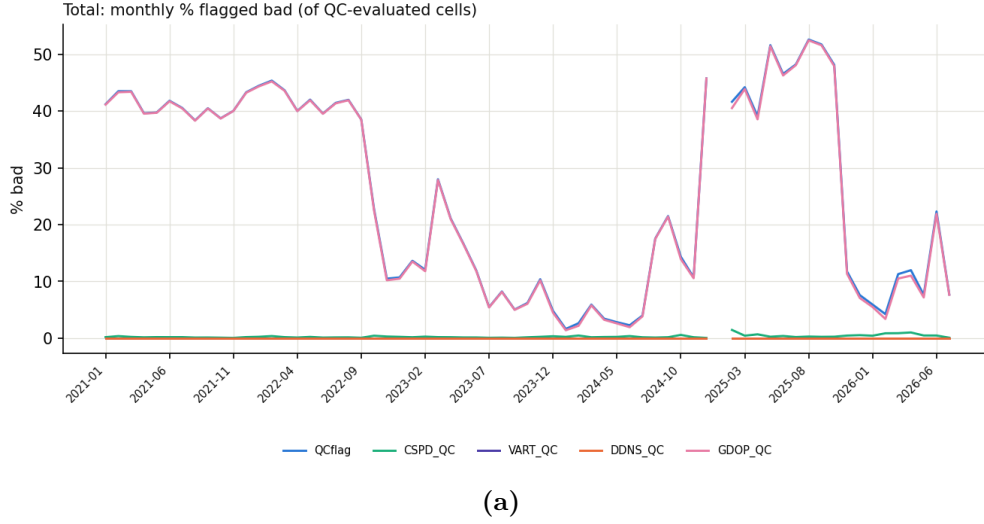


Figure 6: Quality-control results for Total product. (a) Monthly fraction of data flagged as bad by the QC tests. (b) Spatial distribution of the fraction of data flagged as bad over the full observation period.

3.2 Discussion

Radial product:

Taking into account the work carried out under *Section 1* of the project **Quality assessment of surface current and wave measurements from the HFR System in the GOT** (see Figure 3 in *wp1*), and the results just obtained for stations AURI, PIRA, and TRI1, the analysis indicates that the RDCT_QC monthly time series are not noise, in fact they line up with already-known station outages :

- **AURI:** in Figure 2 (a), RDCT_QC climbs sharply from mid-2024, reaching almost 40% in September 2024, immediately before the 14-month coverage gap (September 2024-November 2025).
Spatially, Figure 2 (b), the RDCT_QC/QCflag failure is a small, compact patch near the top of the domain (close to the station itself). The patch is visually distinct from CSPD_QC/VART_QC/MDFL_QC, which show the ordinary diffuse range-decay pattern spreading toward the west/east edges instead.
- **PIRA:** there is a sharp spike around February 2026 in Figure 4 (a), which corresponds to the short outage window at the beginning of the year. This month has very few QC-evaluated cells, so the spike reflects a handful of straggler cells rather than substantial reporting-but-rejected data.
Spatially, Figure 4 (b), PIRA has two separate problem zones: RDCT_QC/QCflag in the south, and CSPD_QC/VART_QC elevated in the north.
- **TRI1:** episodic spikes to around 20-30% in February/April 2024 in Figure 5 (a); otherwise, values are low, matching the intermittent-return pattern from the coverage analysis.
Spatially, Figure 5 (b), RDCT_QC/QCflag failure is broad, covering most of the domain, unlike CSPD_QC/VART_QC, which are concentrated narrowly at the west edge and one SE corner cell.

Moreover, **PIRA** and **IZOL** show also a different pattern: sustained periods where the station was still substantially reporting (QC-evaluated, not ‘no_data’) but RDCT_QC rejected most of it. This is not visible in Figure 3 coverage graph from *wp1*, which only tracks presence/absence of data, not whether it passes QC.

- **IZOL:** in Figure 3 (a) it is visible that RDCT_QC/QCflag sit at around 100% bad continuously from March 2025 through June 2026, then drop sharply back to near 0%. This represents over a year during which IZOL was technically reporting, but the existing QC was correctly rejecting nearly all of the data.
Spatially, we see that in Figure 3 (b), the RDCT_QC failure is concentrated toward the SE portion of IZOL’s coverage domain, reaching > 30% even in the full-record average.
- **PIRA:** Figure 4 (a) shows a recurring winter seasonal degradation, roughly between January and April each year and peaking in February.

Total product: The GDOP_QC accounts for almost all of QCflag’s 18.8% overall bad rate. This can be seen in the spatial map (Figure 6 (b)), which shows that this is not scattered noise: it is a sharp, persistent band running roughly N-S through the middle of the domain ($\sim 13.6 - 13.7^\circ\text{E}$), reaching $> 50\%$ bad in places. We can deduce that this is the geometric-dilution “seam” between the AURI/PIRA baseline, where radial intersection angles are too acute.

3.3 Conclusions

The RDCT_QC and GDOP_QC flags are catching real, substantial degradation events, not just theoretical edge cases.

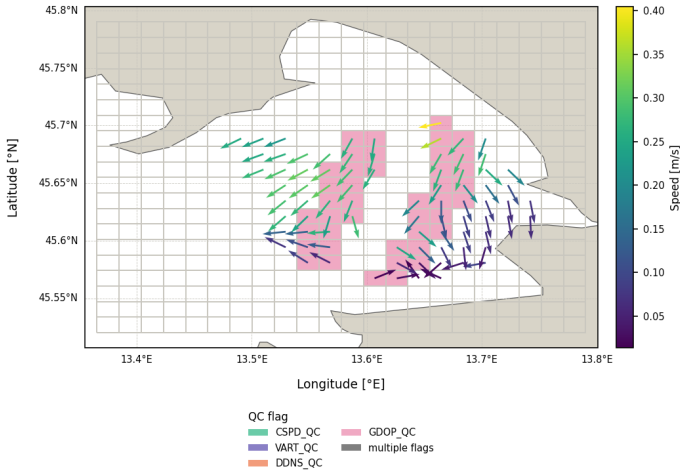
The RDCT_QC and GDOP_QC flags only catch geometric/count problems, not the physical plausibility of the vectors that do pass.

A vector could pass RDCT_QC/GDOP_QC/CSPD_QC/VART_QC and still be a spurious spike or unphysical divergence.

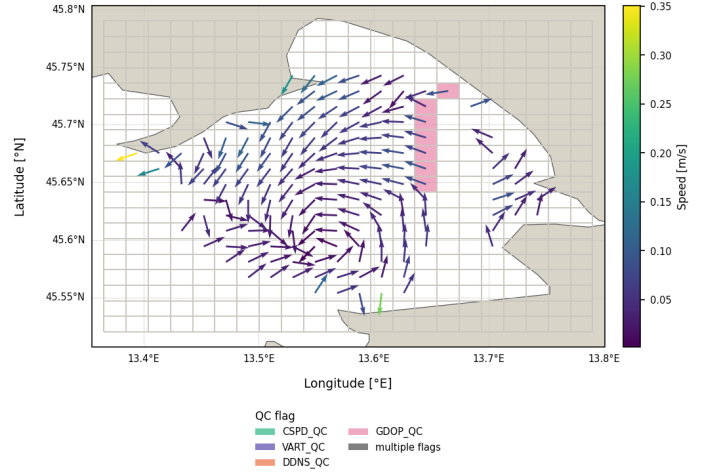
The GDOP “seam” is a ready-made validation case for Sklop 3: any new divergence/consistency test should be checked against this exact zone to see whether it is rediscovering the same geometry-tied region that GDOP_QC already flags, or something new.

4 Identification of known artifacts and anomalies (e.g. from metadata, incidents, visual inspections, or comparisons with independent datasets), and verification of whether they are detected by the existing algorithms

Prior to conducting the analysis and computations, some examples of artefacts observed in the graphical representations are presented. All figures presented herein were generated by the author.



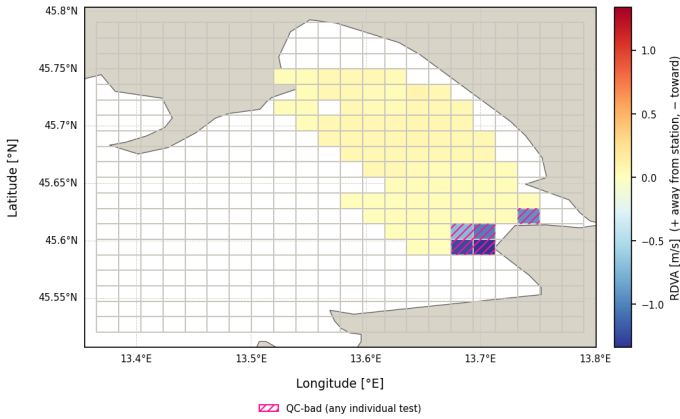
(a) Near Real Time Surface Ocean Total Velocity by HFR-NAdr-Total at 2021-01-01 T00:00:00Z



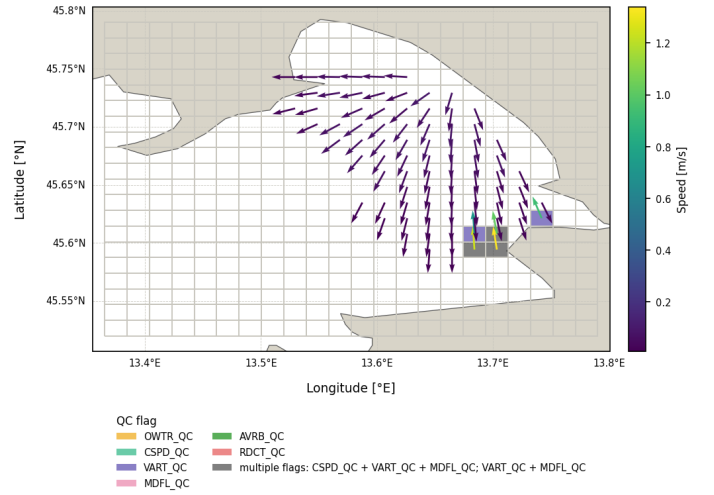
(b) Near Real Time Surface Ocean Total Velocity by HFR-NAdr-Total at 2022-10-21 T06:00:00Z

Figure 7: Examples of surface water velocity artefacts in the GOT.

As can be seen in both Figures 7 (a) and (b), there is a band of missing cells running N-S through the middle of the domain. This is the same GDOP geometric-dilution seam quantified in Task 2 (Figure 6, (b)). Seeing it directly in the current field, independent of the aforementioned QC scripts, confirms it is a real, persistent geometric artefact of the baseline geometry, not a computation error.



(a) RDVA



(b) Velocity vectors

Figure 8: Radial velocity example at HFR-NAdr-AURI, 2022-01-02 T10:30:00Z: (a) RDVA, (b) velocity vectors.

In Figure 8 (a), a small group of adjacent cells near the SE coast shows an abrupt discontinuity against the surrounding RDVA, with no gradual transition. The corresponding vectors, Figure 8 (b), reverse direction relative to every neighboring vector. Following the examination against the unified dataset, this is one of the few cases where the existing QC does catch the anomaly, with QCflag = 4, driven by VART_QC.

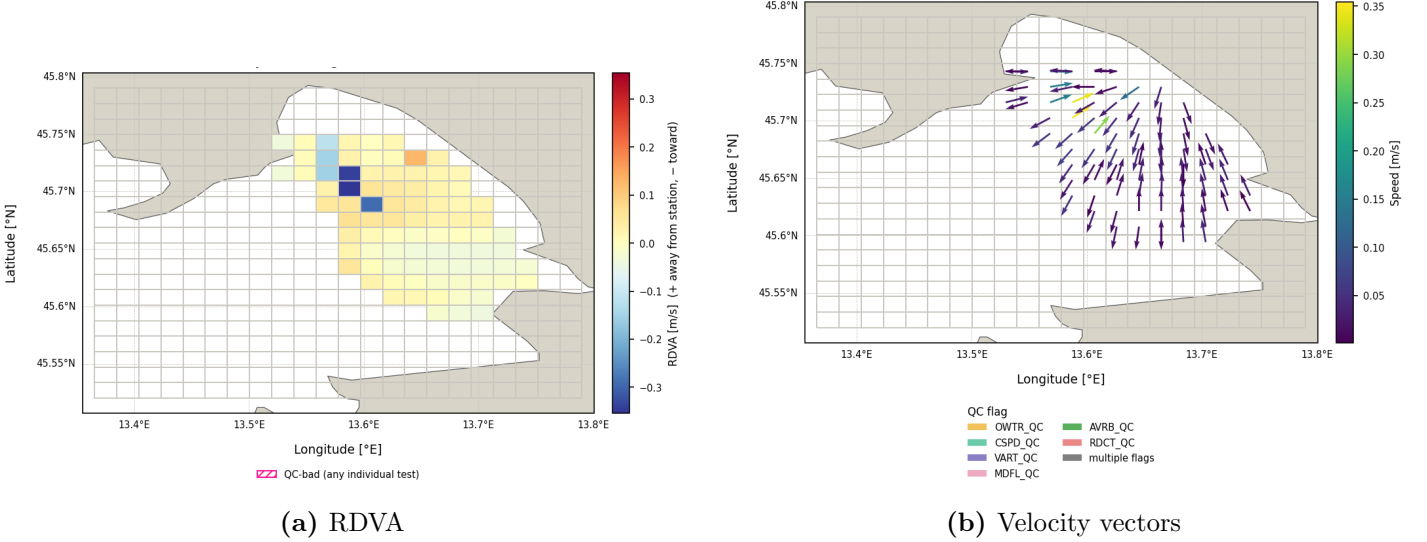
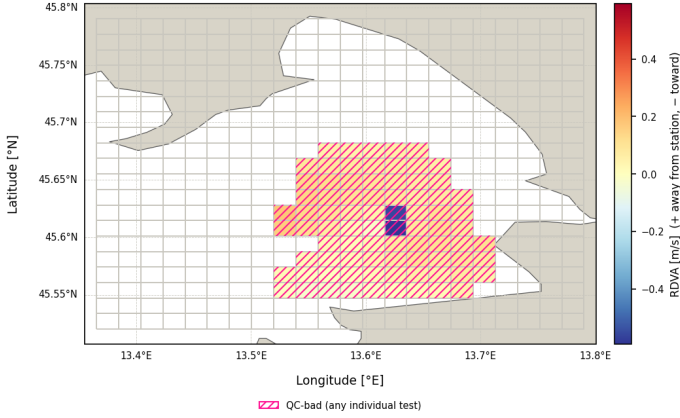


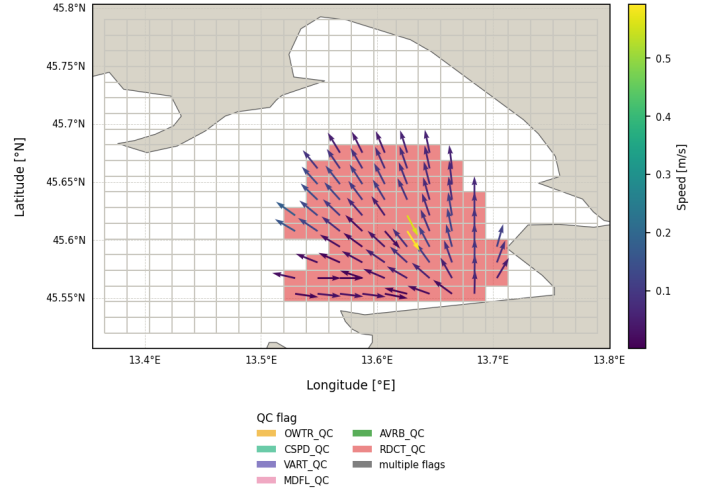
Figure 9: Radial velocity example at HFR-NAdr-AURI, 2022-01-02 T11:00:00Z: (a) RDVA, (b) velocity vectors.

Figure 9 (a) shows a compact cluster of cells, at the AURI antenna site, distinct from the diffuse range-decay pattern shown by the rest of the domain. The vectors in Figure 9 (b) have opposite direction and are higher in magnitude than the surrounding ones. However, at this specific timestamp, every valid cell, including this cluster, carries QCflag = 1: RDCT_QC, OWTR_QC, CSPD_QC, VART_QC and MDLF_QC all pass. This individual event is not caught by any existing test; it only becomes visible once RDCT_QC failures are aggregated across the whole record.

Figure 10 (a) shows one isolated cell dropped into an otherwise smooth field, with no gradient transition on any side. In (b), two vectors cut diagonally across an otherwise uniform field of low-magnitude, consistently-oriented vectors. This cell carries QCflag = 4. However, OWTR_QC, CSPD_QC, VART_QC and MDLF_QC all pass individually here, and the QCflag = 4 comes entirely from RDCT_QC.



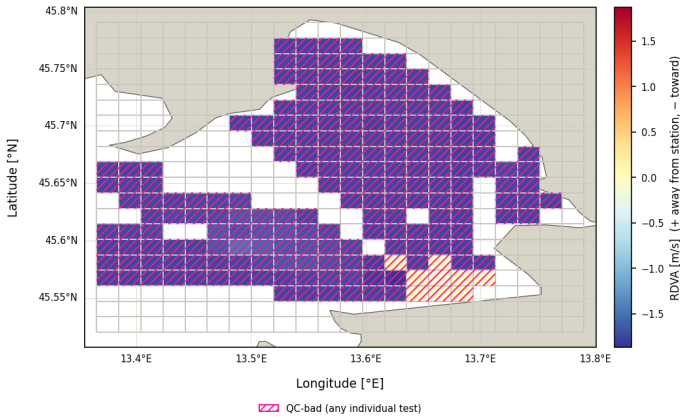
(a) RDVA



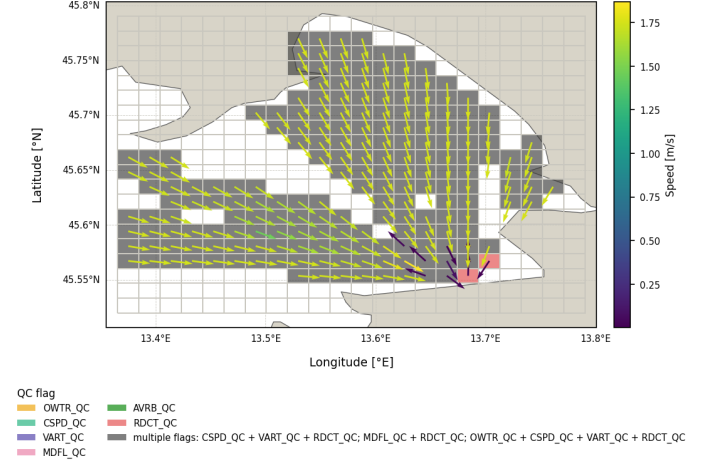
(b) Velocity vectors

Figure 10: Radial velocity example at HFR-NAdr-IZOL, 2025-02-21 T00:00:00Z: (a) RDVA, (b) velocity vectors.

Figure 11 shows both the RDVA field (a) and the velocity-vector field (b) during the IZOL outage window. Nearly the entire domain is flagged, consistent with a whole-scan RDCT_QC rejection rather than a cell-specific anomaly.



(a) RDVA



(b) Velocity vectors

Figure 11: Radial velocity example during outage at HFR-NAdr-IZOL, 2025-12-16 T23:00:00Z: (a) RDVA, (b) velocity vectors.

5 Report on the effectiveness of the existing QC algorithms with recommendations for upgrade (Sklop 3)

Tasks 2 and 3 establish two different failure modes for the existing QC:

- Where an existing test is well-matched to a real problem, RDCT_QC against station outages and low radial count, GDOP_QC against the AURI/PIRA baseline geometry, CSPD_QC/VART_QC/MDFL_QC against extreme or unstable individual vectors - it catches it reliably, and the flagged rate tracks known incidents rather than noise. Task 2’s station-by-station results, and Figures 7, 8, 10, and 11 above, are direct evidence of this.
- Where the failure has no matching existing test, it passes as “good data” regardless of how implausible it looks. Figure 9 (a near-field artefact with all 7 radial tests passing) is a concrete, verified example.

Product / test family	What it reliably catches	Known blind spot	Recommendation for Sklop 3
Radial – RDCT_QC	Station outages and whole-scan low radial count	Cell-specific anomalies inside an otherwise “good” scan (Figure 9)	Add a per-cell consistency/divergence test independent of scan-wide radial count
Radial – CSPD_QC / VART_QC / MDFL_QC	Single vectors that are extreme or temporally unstable (Figure 8)	Vectors wrong in direction/context but not extreme in magnitude	Add a neighbor/bearing-consistency test, that would flag direction rather than only magnitude
Total – GDOP_QC	The AURI/PIRA geometric seam, consistently, ~18.6% of the domain (Task 2)	Non-physical divergence/convergence and direction reversals in the domain	Implementation of divergence/convergence detection tests

Table 5: Summary of existing QC effectiveness (Sklop 2, Tasks 1–3) and corresponding recommendations for Sklop 3.

Overall, the existing EU HFR Node QC does its intended job within its own scope, i.e. geometry- and count-based rejection. However, it was not designed to catch physical-consistency problems: non-physical divergence, direction reversals, and neighbor-inconsistent vectors. That is exactly the gap Sklop 3 is scoped to close.